Gestão de Projecto de Software

Licenciatura em Engenharia Informática

2024/2025

Estimation

A5

172

Estimar

- Após a aprovação das Us pelo PO a ST é responsável pelo processo de estimativa
- Estimar com precisão o tempo e os requisitos a nível de recursos humanos é um factor crítico num projecto para garantir qualidade no Product Increment
- A estimativa tem sempre um erro associado, que é influenciado:
  - complexidade do projecto
  - Tamanho da equipa
  - A experiência da equipa em estimar
- O Scrum procura melhorar a precisão nas estimativas recorrendo a actividades e a métodos de comparação relativos

173

What is estimation?

- The project manager must set expectations about the time required to complete the software among the stakeholders, the team, and the organization's management.
- If those expectations **are not realistic** from the beginning of the project, **the stakeholders will not trust the team or the project manager**.

What is an estimate?

- "A tentative evaluation or rough calculation."
- "A preliminary calculation of the cost of a project. "
- "A judgment based upon one's impressions; opinion. "
  - Source: The American Heritage Dictionary, Second College Edition, 1985
- It's not:
  - A commitment or a plan to meet a target

Why estimate?

- To support
  - Contract negotiation – cost and schedule
  - Project planning
  - Project control

- "... determine whether a project's targets are realistic enough to allow the project to be controlled to meet them."
  - Source: Software Estimation – Demystifying the Black Art, 2006

Como criar uma estimativa consistente/sólida

- Deve existir:
  - Uma estimativa do esforço de cada tarefa
  - É necessário elaborar uma lista de pressupostos em que a estimativa se baseia
  - Deve haver um consenso na equipa sobre a necessidade da exixtência da estimativa e sobre a própria estimativa.

How good an estimator are you?

- For each question, fill in the upper and lower bounds that, in your opinion, give you a 90% chance of including the correct value

- What is the surface temperature of the Sun?
  - Low estimate:
  - High estimate:

- How long does it take to go from ISEC in Coimbra to ISEL in Lisbon?
  - Low estimate:
  - High estimate:

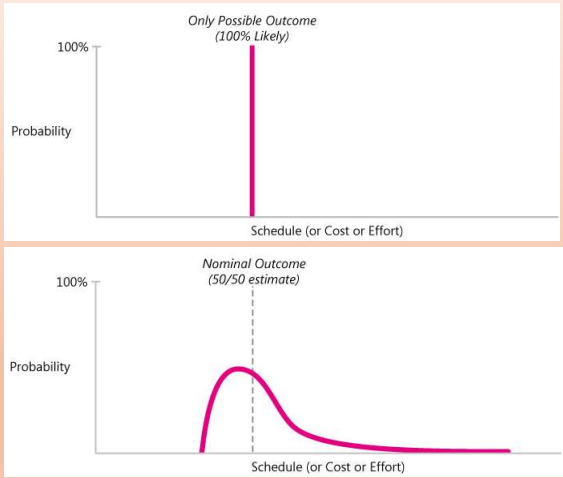
Influencing factors

- Which factors did you took in consideration?
  - Confidence level
  - Knowledge base/ historical data
  - Personal experience
- For the travelling time estimation
  - Transport (train, car, bus, ...)
  - Route (highways, secondary roads, ...)
  - Traffic hours
  - Complying with driving rules (speed limit, ...)
  - Resting/eating pauses
  - Unexpected conditions (weather, accidents, ...)
- Did you make any assumptions?
- Did you consider several options?

Assumptions Make Estimates More Accurate

- Team members make assumptions about the work to be done in order to deal with incomplete information
  - Any estimate must be based on a decision that has not yet been made; team members can assume the answer for the sake of the estimate
  - Assumptions must be written down so that if they prove to be incorrect and cause the estimate to be inaccurate, everyone understands what happened
  - Assumptions bring the team together very early on the project so they can make progress on important decisions that will affect development

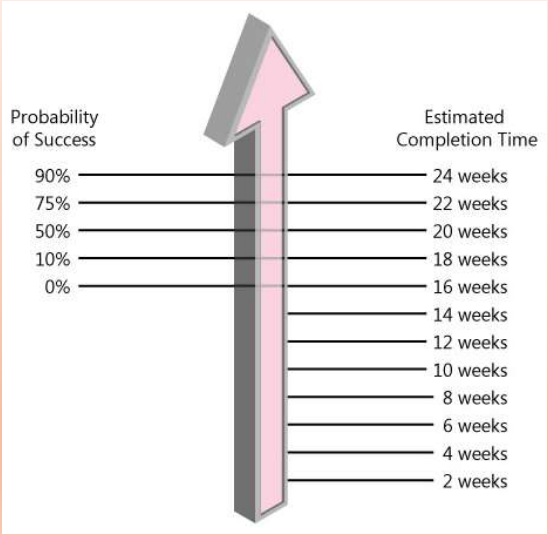
Estimation as probability statements



Single-point estimates assume 100% probability of the actual outcome equaling the planned outcome **UNREALISTIC!**

Project outcomes: There is a limit to how well a project can go but no limit to how many problems can occur **REALISTIC!**

Estimation as probability statements



182

Main issues when estimating

- Complexity of systems
  - Non-repeatable work
- Bias torward underestimation
  - When overestimation occurs there's Parkinson's Law
- Estimates are in fact goals
  - When estimating we manipulate the output to achive the goal
- Inaccurate information about organization capabilities
  - Missing or unused historical data
- Unclear requirements
  - Stakeholders with different understanding of what to develop
  - Hidden assumptions

- Tendência para sub-estimar a eficiência necessária para a realização de uma tarefa e hiper-estimar o tempo que necessita para realizar a tarefa.
- Tendência humana de ocupar mais tempo do que o necessário para realizar uma tarefa quando existem prazos muitos longo (com consequência perda de eficiência)

183

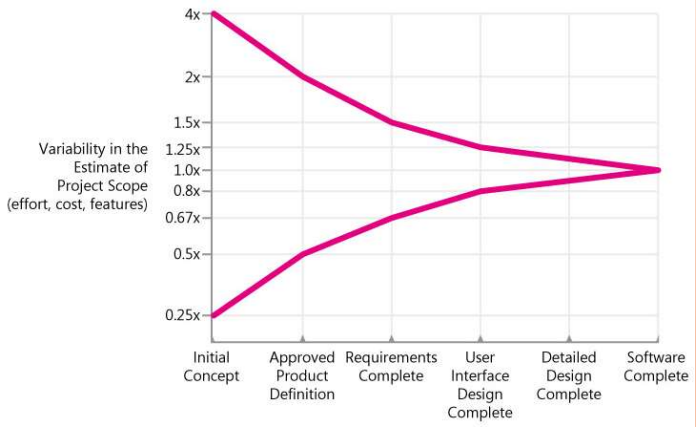
WBS – work breakdown structute

- Need to decompose the project into manageable chunks
- Hierarchical list of project work activities

- Ex:
  - User Story 1
    - Task11
    - Task12
  - User Story 2
    - Task21
    - Task22

When to estimate?

- The Cone of Uncertainty



Source: *Software Estimation – Demystifying the Black Art*, 2006

How to estimate?

- Estimation techniques
  - Wild Altogether Guess
  - Experience - experts judgment
  - Wideband Delphi
    - experts judgement in groups
    - Variants: Estimation game, Planning poker
  - Estimation by Analogy
  - Function Points Analysis
    - Variants: Use Case Points, Object Points, COSMIC, ...
  - COCOMO II
  - Bottom-up
    - estimate each component
  - ...

Método Delphi

- método de Delphi, ou o painel de Delphi, é um processo estruturado, que visa a obtenção de um consenso sobre um determinado assunto, com base na opinião de um grupo de peritos.
- foi criado na década de 50 pela firma Rand Corporation (Santa Monica, California). Numa fase inicial destinava-se ao uso militar (nomeadamente pela *Americam Armed Forces*), tendo sido permitido o uso civil na década de 60

What and how to estimate in agile?

- Product Backlog -> User Stories -> Story Points
- Sprint Backlog -> Tasks -> Hours/Days

Story points

- Story points are a unit of measure to estimate the effort required to complete a specific task, user story, or feature.
- They are a **relative measure** of complexity and effort rather than a precise measure of time.
- Help development teams plan and prioritize their work more effectively by providing a common understanding of the work involved.
- Compare items based on:
  - **risk** or **uncertainty** associated with the tasks
  - the **amount of work** from team's experience with similar tasks
  - the tasks **complexity**
- Estimate the story points using e.g. **Planning Poker**, based on the Fibonacci sequence (0, 0.5, 1, 2, 3, 5, 8, 13, 20, 40, 100), or **T-Shirt Size**

Sequência de números na qual cada termo subsequente corresponde à soma dos dois anteriores

$$F(n) = F(n-1) + F(n-2)$$

Condições iniciais:

$$F(0) = 0$$
$$F(1) = 1$$

188

How to estimate – Planning Poker



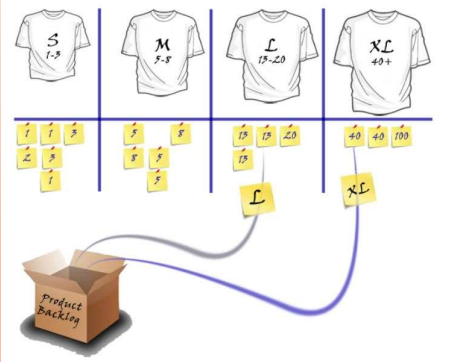
189

How to estimate – Planning Poker

1. Form a group of no more than 10 estimators and a moderator. The product owner or client manager cannot be an estimator.
2. Each estimator gets a deck of cards: 0, 0.5, 1, 2, 3, 5, 8, 13, 20, 40, 100 and ?.
3. The moderator reads the description of the user story or task. The product owner answers questions from the estimators.
4. Every estimator selects an estimate card and places it face down on the table. After all estimates are in, the cards are flipped over.
5. If the estimates vary widely, the owners of the high and low estimates discuss the reasons for diverging. All estimators should participate in the discussion, 3 minutes max.
6. Repeat from step 4 until the estimates converge.

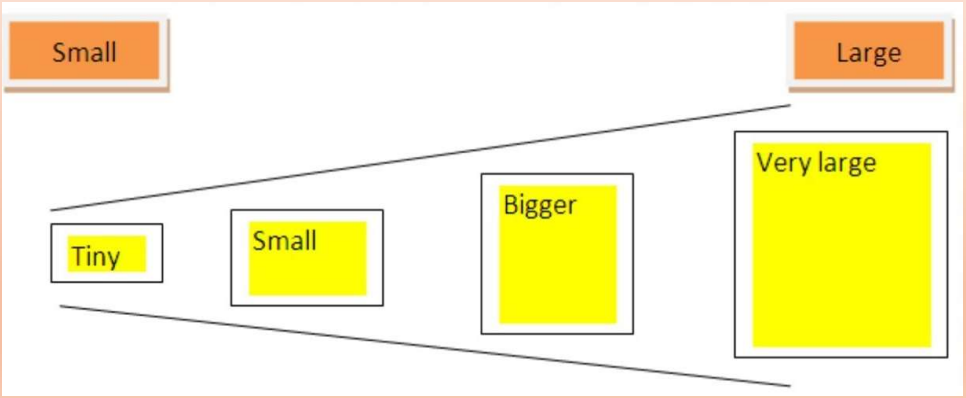
Estimate with T-Shirt Size

- Relative estimation technique, usually applied to user stories
- In planning poker, rather than using a number, Items are classified into t-shirt sizes, ex: XS, S, M, L, XL.



- Em vez de se tentar calcular o número exacto do que se pretende avaliar, avalia-se a dificuldade, a importância, o "tamanho" … de forma relativa
- Vantagens:
  - Simplicidade
  - Rapidez
  - Facilidade de compreensão
    - Pessoas de diferentes áreas podem facilmente entender o conceito dos tamanhos.
  - Estimativa relativa

Affinity Estimating



192

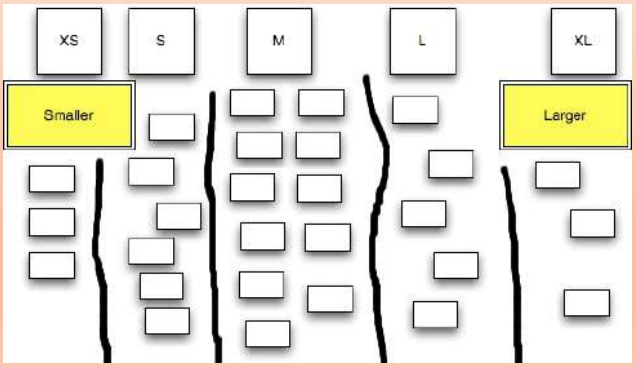
How to estimate – Affinity Estimating

1. Place two posts on the Wall (small and large).
2. Team members will be asked to come to the Wall with a subset of the product backlog.
3. Team members will size each item relatively to the others on the wall.
   - Silent step.
4. Product owner/Stackeholders present only to provide clarity on some issue.
5. Place all the items on the Wall.
6. After that, all members would look to the Wall and validate if they agree on the estimation.
7. They may change the place of the user story after some debate.
8. Estimation ends when no more user stories are changed in the Wall.

193

How to estimate – Affinity Estimating

9. Create columns
   - Relative buckets.
   - T-shirt or fibonacci sequence



Prioritisation with MoSCoW

- In fixed-time projects it is vital to understand the relative importance of the work to be done
- MoSCoW helps to understand and manage priorities, mainly in User Stories

- Must Have
  - there is no point in implementing a solution that does not meet this requirement
- Should Have
  - important but not vital, although painful to leave out, the solution is still viable
- Could Have
  - wanted or desirable but less important, has less impact than "should have" if left out
- Won't Have this time
  - not be delivered for now, maybe later; helps keeping the focus at this timeframe on the more important requirements

Exemplo de aplicação do MoSCoW

Estamos a desenvolver uma aplicação de gestão de tarefas

- priorizar as *funcionalidades* desenvolver
  - **Must have:**
    - Funcionalidade de criar, editar e eliminar tarefas (sem isto, a aplicação não seria funcional).
  - **Should have:**
    - Notificações de prazos (importantes, mas a aplicação ainda pode funcionar sem elas).
  - **Could have:**
    - Integração com calendários externos (seria útil, mas não é fundamental para o uso básico da aplicação).
  - **Won't have this time:**
    - Funcionalidade de atribuição de tarefas a múltiplos utilizadores (não será feita nesta versão, mas pode ser introduzida no futuro).

196

196

Vantagens do método MoSCoW

- Clareza
  - uma forma clara e fácil de comunicar prioridades dentro da equipa e com as partes interessadas.
- Foco
  - ajuda a focar recursos e tempo nas tarefas mais críticas, garantindo que os requisitos essenciais são cumpridos primeiro.
- Flexibilidade
  - adapta-se bem a projectos ágeis, permitindo que funcionalidades menos prioritárias sejam adiadas para futuras versões sem comprometer o projecto actual.

197

197

Further reading

- Andrew Stellman and Jennifer Greene, *Applied Software Project Management*, O'Reilly, 2006. [Section 3]
- Whitney Vige, What are story points? Six easy steps to estimate work in Agile, December 3rd, 2022 (https://asana.com/resources/story-points)
- Mountain Goat Software website, *Planning Poker*, https://www.mountaingoatsoftware.com/agile/planning-poker
- Many online tools (ex: https://planningpokeronline.com/)

198

Gestão de Projecto de Software

Licenciatura em Engenharia Informática

2024/2025

199